

Can an HTTP response message contain an empty body? If yes, when can this happen?

Options:

- A) Yes, if the requested object is recently modified.
- B) Yes, if the requested object has been moved to a new location.
- C) No, an HTTP response message can never have an empty body.
- D) Yes, if the requested object is very small in size.

Answer:

- A) Yes, if the requested object is recently modified.

An HTTP response can have an empty body in certain conditions, such as when the requested object is modified recently, or if there's no content to return (e.g., status code 204 No Content).

2. Using Conditional GET in the HTTP Protocol, the number of RTT (Round Trip Time) is reduced if the object has not been modified since the date of the cached copy!

Options:

- A) False
- B) True

Answer:

- B) True.

Using Conditional GET reduces the number of RTTs when the cached object has not changed, as the server will return a 304 Not Modified response.

3. Top-level domain (TLD) servers are an organization's own DNS servers, providing authoritative hostname to IP mappings for the organization's named hosts.

Options:

- A) Yes
- B) No

Answer:

- B) No.

TLD servers are not specific to any one organization. They handle the authoritative domain name resolution for the top-level domains like .com, .org, etc., but not the organization's internal host mappings.

4. How many sockets are there in a TCP server communicating with 13 clients concurrently?

Options:

- A) 13 sockets
- B) 14 sockets

- C) 1 socket
- D) None of the options

Answer:

- B) 14 sockets.

The server uses one socket for listening to incoming connections, and one socket for each of the 13 clients, so there are 14 sockets in total.

5. Which of the following statements about DNS is TRUE?

Options:

- A) TLD servers store all the hostname-to-IP mappings of the Internet.
- B) DNS servers listen to TCP port 53.
- C) Every Web server must have a canonical name.
- D) The local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.

Answer:

- D) The local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.

Local DNS servers can occasionally provide outdated mappings due to cached data.

6. A source and a destination are separated by 300,000 kilometers and are connected by a direct link with a speed of 2 Kbps. The propagation speed over the link is 20,000 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is 2×10^4 bits long. Suppose ACK packets are of negligible size and the transmission channel is error-free. What will be the transmission time?

Options:

- A) 5 seconds
- B) 100 seconds
- C) It depends on the propagation delay
- D) Transmission time cannot be calculated

Answer:

- C) It depends on the propagation delay.

The transmission time is calculated by considering both the time taken to send the packets and the propagation delay over the distance. Both factors need to be combined.

7. What happens if the packets are too small in size for HTTP or TCP protocols?

Options:

- A) The packets are lost due to insufficient bandwidth.

- B) The network load increases due to an increased number of packets.
- C) The protocols automatically adjust the packet size.
- D) It has no impact on the network.

Answer:

- B) The network load increases due to an increased number of packets.